

- 1 Tick (✓) **one** box to show which word accurately describes the scope of a variable declared in a procedure.

A	Function	☐
B	Global	☐
C	Local	☐
D	Subroutine	☐

[1]

- 4 Describe how variables and constants are used in programming.

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..... [3]

10 The variables X, Y and Z are used in a program: X stores a whole number, Y stores a decimal number and Z stores a flag that can be set to TRUE or FALSE

(a) Write pseudocode statements to declare the variables X, Y and Z

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..... [3]

(b) The function Same (A, B) returns TRUE if the value of A is the same as the value of B when B is rounded to the nearest whole number and FALSE otherwise.

- Write pseudocode statements to:
- define the function
 - call the function with X and Y and store the return value in Z

Function definition

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Function call

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(c) State the difference between defining and calling a function.

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..... [1]

- 2 Tick (✓) **one** box to show the name of the data structure used to store a collection of data of the same data type.

A Array

☐

B Constant

☐

C Function

☐

D Variable

☐

[1]

- 4 Explain the purpose of the library routines `DIV` and `ROUND`

`DIV`

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`ROUND`

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[4]

- 6 State **two** features that should be included to create a maintainable program.

Give a reason why each feature should be used.

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[4]

8 The function `LENGTH (Phrase)` calculates the length of a string `Phrase`

- (a) Write the pseudocode statements to:
- store the string `"The beginning is the most important part"` in `Phrase`
 - calculate and output the length of the string
 - output the string in upper case.

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- (b) Write the output your pseudocode should produce.

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- 6 State **three** different features of a high-level programming language that a programmer could use to make sure that their program will be easier to understand by another programmer. Give an example for each feature.

Feature 1

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Example

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Feature 2

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Example

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Feature 3

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Example

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[6]

11 The variables `P` and `Q` are used to store data in a program. `P` stores a string. `Q` stores a character.

- (a)** Write pseudocode statements to declare the variables `P` and `Q`, store "The world" in `P` and store 'W' in `Q`

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- (b)** Write a pseudocode algorithm to:

- convert `P` to upper case
- find the position of `Q` in the string `P` (the first character in this string is in position 1)
- store the position of `Q` in the variable `Position`

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- (c)** Give the value of `Position` after the algorithm has been executed with the data in question 11(a).

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- 1 Tick (✓) **one** box to complete the sentence.

A constant

- A** stores a value that can change at any time during the execution of a program.
- B** stores a value that cannot change during the execution of a program.
- C** stores values of multiple data types.
- D** stores values that must be of the same data type.

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[1]

- 2 Explain the purpose of the library routines MOD and RANDOM

MOD

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RANDOM

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[4]

- 3 Describe what happens when a function is called during the execution of a program.

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[3]

9 The variable `Saying` is used to store string data in a program.

(a) Write the pseudocode statement to declare the variable `Saying`

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(b) Write the pseudocode statements to:

- allow a string to be input to the variable `Saying`
- store the content of the variable `Saying` in a text file named `"Quotations.txt"`
- make sure the text file is closed at the end of the algorithm.

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..... [5]

- 1 Tick (✓) **one** box to show which operator means less than or equal to.

A	OR	☐
B	<	☐
C	<=	☐
D	>=	☐

[1]

- 2 Tick (✓) **one** box to show how a value can be passed to a procedure.

A	function	☐
B	parameter	☐
C	return	☐
D	subroutine	☐

[1]

- 3 **Four** descriptions of data and **five** data types are shown.

Draw **one** line to link each description to the most appropriate data type.

Not all data types will be used.

Description	Data type
a whole number	BOOLEAN
a single letter	CHAR
a word or phrase	INTEGER
a number with two decimal places	REAL
	STRING

[4]

- 5 Explain how variables and constants should be used when creating and running a program.

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..... [3]

7 The string operation SUBSTRING(Quote, Start, Number) returns a string from Quote beginning at position Start that is Number characters long. The first character in Quote is in position 1.

Write pseudocode statements to:

- store the string "Learning Never Exhausts The Mind" in Quote
- extract and display the words "The Mind" from the string
- output the original string in lower case.

[5]

8 Explain why a programmer would use procedures and parameters when writing a program.

[4]



2 Tick (✓) **one** box to show which library routine returns the remainder of a division.

A DIV

☐

B MOD

☐

C RANDOM

☐

D ROUND

☐

[1]

7 A program uses both local variables and global variables.

Describe **two** differences between local variables and global variables.

Difference 1

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Difference 2

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[4]